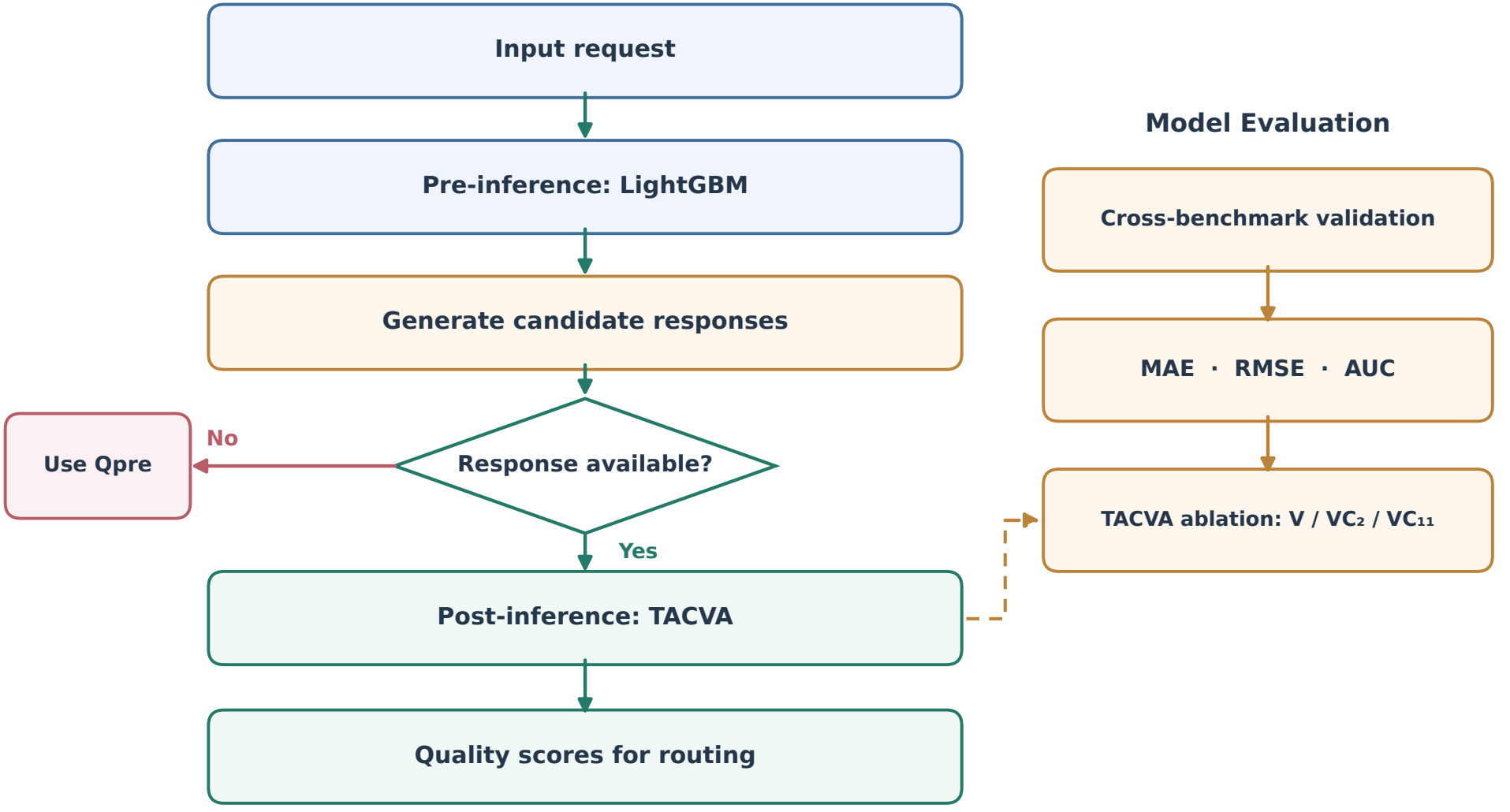
Input request
Pre-inference: LightGBM
Generate candidate responses
Response available?
No
Use Qpre
Yes
Post-inference: TACVA
Quality scores for routing
Model Evaluation
Cross-benchmark validation
MAE · RMSE · AUC
TACVA ablation: V / VC2 / VC11